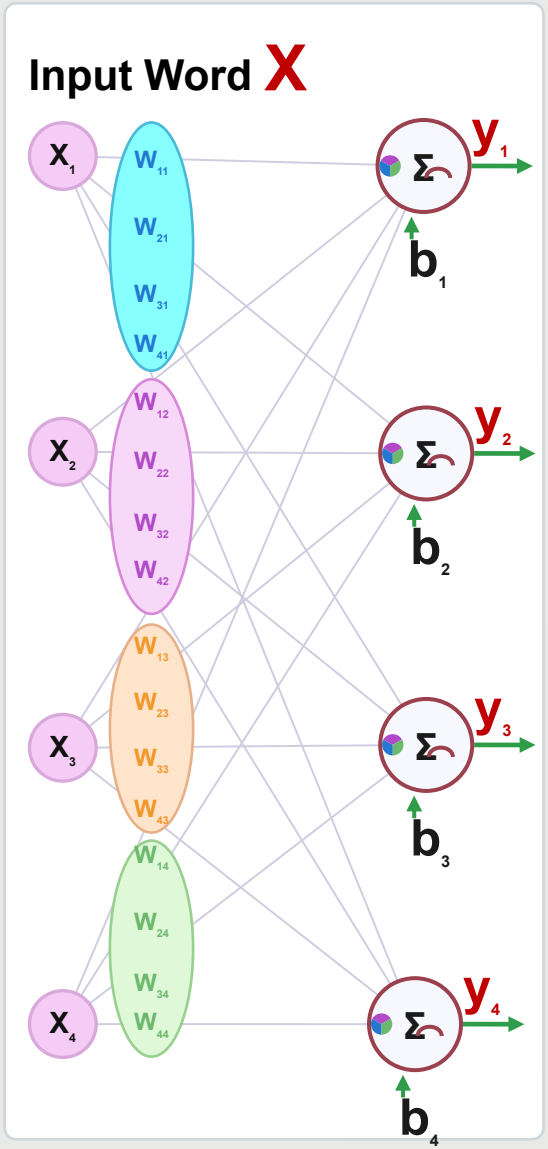
# Using FF NN for Multiplication of a Vector by a Matrix




$$\begin{bmatrix} y_1 & y_2 & y_3 & y_4 \end{bmatrix}^T = Sigmoid\left( \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix} X \begin{bmatrix} w_{11} & w_{21} & w_{31} & w_{41} \\ w_{12} & w_{22} & w_{32} & w_{42} \\ w_{13} & w_{23} & w_{33} & w_{43} \\ w_{14} & w_{24} & w_{34} & w_{44} \end{bmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix} \right)$$